

# *A Novel design of a high speed hysteresis-based comparator in 90-nm CMOS technology*

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**Abstract**— Analog-to-digital circuit converts an analog signal having continuous-time and continuous-amplitude to a discrete-time and discrete-amplitude digital signal. A comparator is a vital block in any analog-to-digital circuit. The comparators play a crucial part in the analog to digital conversion. This paper puts forth the design of hysteresis comparator in 90-nm CMOS technology. The proposed comparator reduces the occurrence of noisy output and has high speed, in comparison to the conventional comparator. The circuit design and analysis has been done using Cadence.

**Keywords**—comparator; noisy output; analog-to-digital conversion; hysteresis; noisy comparator

## I. INTRODUCTION

A comparator performs the comparison of an analog signal with a reference signal or any other analog signal, and gives the output of the comparison in form of a binary signal. In a simplified tone, a comparator can be termed as a one-bit analog-to-digital converter (ADC). In the areas of analog to digital conversion, comparators are predominantly used. Comparators are also widely employed in various other electronics devices as well, like the null detector, zero crossing detector, relaxation oscillator, level shifter and window detector.

In this world of miniaturization, technology is being scaled down. Designs of circuits with scaled down technology is extremely essential. This paper thus presents the design of hysteresis-based comparator in 90-nm CMOS technology. These comparators are preferred over other existing or conventional circuits, as they eradicate the occurrence of a noisy output, in case of operation of the comparator in a noisy environment. It also exhibits high speed.

Section II deals with the basic details of the comparator, Section III puts forth the proposed circuit design and the operation. Section IV deals with the results and Section V concludes the paper.

## II. BACKGROUND

As we know, a comparator performs the comparison operation of two analog signals and outputs a certain digital output, based on the comparison. There are two types of comparators: open-loop and regenerative comparators. The op-amps without compensation are called the open-loop comparators and the ones that have positive feedback are the regenerative comparators. There is also another type of comparator which is a combination of the two. This comparator is extremely fast.

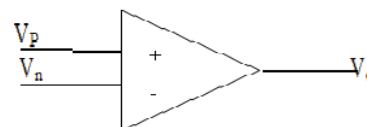


Fig.1. Block diagram of a Comparator

Fig. 1 shows the block diagram of a comparator. When the difference between the inverting and non-inverting inputs is positive, the comparator output is also high and when the difference is negative, the comparator output is low.  $V_{OH}$  and  $V_{OL}$  are the upper and lower voltage limits of the comparator output respectively. [1]

The output of a comparator will for sure be irregular, when the input contains large amount of noise. The noise is an unwanted signal, which is in no way related to the input signal. The noisy inputs can be prevented by using hysteresis-based comparators.

In the hysteresis-based comparators, the input voltage is applied to the inverting input. The input threshold changes as a function of the input. The input threshold voltage is reduced whenever the input passes the threshold.

Dynamic latches have high speed and thus are generally preferred in designs of comparators. There has been a lot of work on design of dynamic latches. An offset measurement and cancellation technique for dynamic latches was put forth.

[2]. There has also been design on a mismatch insensitive dynamic comparator [3]. There also has been work on hysteresis-based comparator, but this paper shows the design and the design results in scaled down technology.

### III. PROPOSED DESIGN

Generally, a comparator is expected to detect signal transitions at the threshold point, when placed in a noisy environment. If a fast comparator (frequency of operation of comparator in comparison to the frequency of the most prevalent noise) is considered and even the amplitude of the noise is great enough, the output will also be noisy. In such a situation, the transfer characteristics of the comparator become desirable to be modified. To be specific, hysteresis is essential to be introduced in the comparator. “Hysteresis is the quality of the comparator in which the input threshold changes as a function of the input (or output) level”.

The input threshold voltage is reduced whenever the input passes the threshold, so that the input reaches a point beyond the previous threshold, prior to the change of the comparator's output state. Though the input changes from negative to positive, the output doesn't change until it reaches the positive trip point ( $V_{TRP+}$ ). But, the effective trip point is changed, once the output goes high. Similarly, when the input returns to its original direction (negative), the output does not change until it reaches the negative trip point ( $V_{TRP-}$ ). This operation is shown in Fig.2.

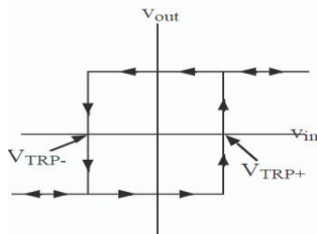


Fig. 2. Figure showing the positive and negative trip points of operation in hysteresis based comparator

The advantage of introducing hysteresis in a noisy environment is shown in Fig. 3. A noisy signal is given as the input to a basic comparator (without hysteresis). The aim is to achieve the comparator following low-frequency signal. The comparator output is noisy, as there are a lot of noise variations near the threshold points. The output response of the comparator can be improved by adding hysteresis, which in amplitude is either equal to or greater than the amplitude of the largest expected noise.

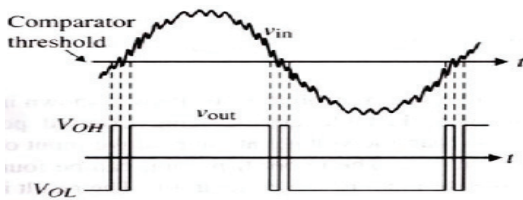


Fig. 3(a). Output of a basic comparator (without hysteresis)

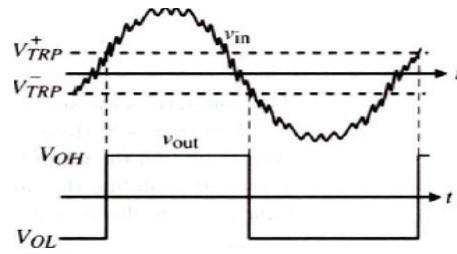


Fig. 3(b). Output of hysteresis based comparator

Fig. 3(a) shows the output response, of a basic comparator without hysteresis, to a noisy input and Fig. 3(b) shows the output response to a noisy input when hysteresis is added (i.e. for a hysteresis-based comparator).

Hysteresis can be introduced in a comparator in various ways (external methods and internal methods). Generally, all of the methods employ positive feedback. External method involves introducing external hysteresis using external positive feedback. The external method can be done after the comparator is built. On the other hand, internal hysteresis does not require an external feedback and is realized internally.

The figure below (Fig. 4) shows the differential input stage of a comparator. In this circuit there are two paths of feedback.

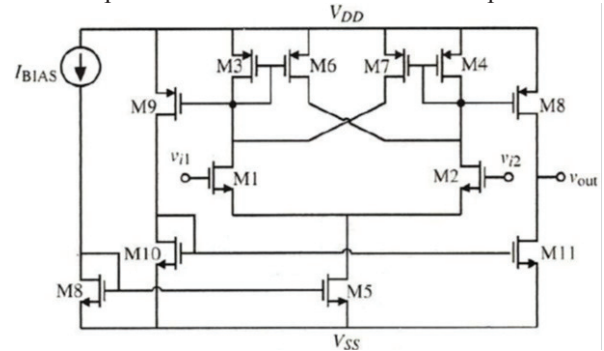


Fig. 4. Comparator with internal hysteresis including an output Stage

The first feedback path is a negative feedback, through the common-source node of M1 and M2 (the current-series feedback). The second feedback path is the positive feedback, through the gate-drain connections of M6 and M7 (the voltage shunt feedback). The feedback factor should be positive for hysteresis to occur; i.e. the positive feedback factor should be greater than the negative feedback factor. Hysteresis comes into picture in the transfer function, only if the ratio  $\beta_6/\beta_3$  is greater than one.

### IV. SIMULATION RESULTS

The circuits are designed and the analysis is done using Cadence Virtuoso in 90-nm CMOS technology.

The designed schematic of the latch-based comparator is shown in Fig. 5. The design is done in the analog environment of Cadence, i.e. Cadence Virtuoso, selecting gpdk 45 library. The circuit was designed and then simulation was done.

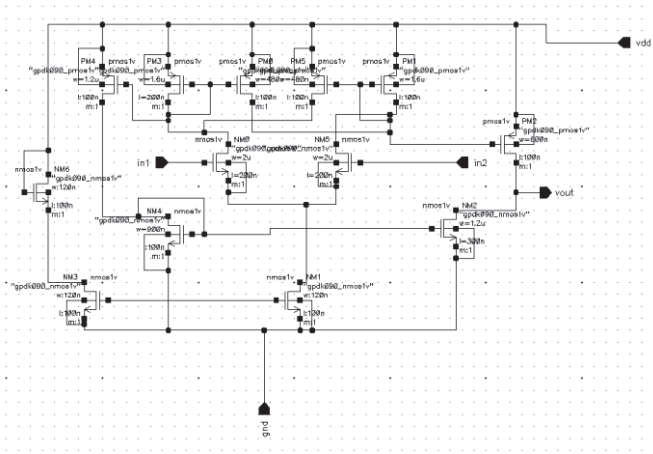


Fig. 5. Comparator with Internal Hysteresis

The transient response of the comparator without hysteresis is shown in Fig. 6(a) and comparator with hysteresis is shown in Fig. 6(b). It is evident that the hysteresis-based comparator output has less noise in comparison to the conventional comparator.

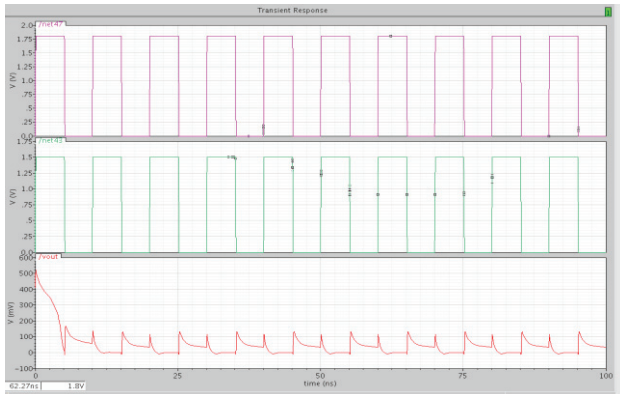


Fig. 6(a). Transient Response of a basic comparator

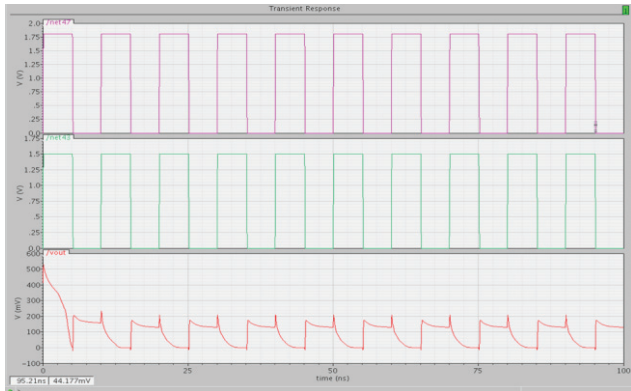


Fig. 6(b). Transient Response of comparator with hysteresis

The layout of the proposed circuit is shown in Fig. 7(a) and the ‘av extracted’ view in Fig. 7(b).

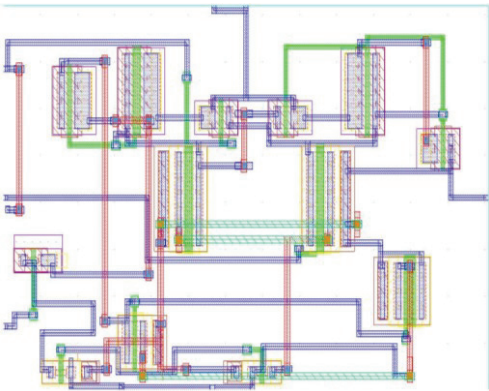


Fig.7(a). Layout of the Hysteresis-based comparator

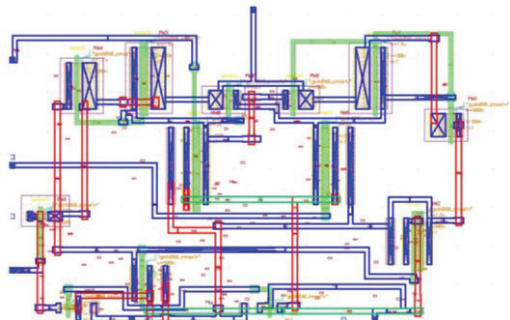


Fig. 7(b). av Extracted view of Hysteresis based Comparator

Table I shows the results of the discussed comparator in view of various parameters. The speed of the circuit is very high. This is because the hysteresis-based comparator output has less glitch and less delay in the comparator components, in comparison to a conventional three stage comparator.

Table II puts forth the same results of the hysteresis-based comparator (as in Table I) in comparison with the conventional 3-stage comparator. The values have been found from the simulation results. The comparison is plotted in Fig. 8.

TABLE I. PERFORMANCE PARAMETERS

| Parameters        | Value                   |
|-------------------|-------------------------|
| Speed             | 9.28GHz                 |
| Delay             | 0.435ns                 |
| Area              | 197.134 $\mu\text{m}^2$ |
| Power dissipation | 2.09mW                  |

TABLE II. PERFORMANCE PARAMETERS COMPARISON

| Comparator name                     | Delay (ns) | Speed (GHz) | Power Dissipation (mW) | Area ( $\mu\text{m}^2$ ) |
|-------------------------------------|------------|-------------|------------------------|--------------------------|
| conventional three stage comparator | 0.168      | 5.59        | 0.1034                 | 197.08                   |
| comparator with hysteresis          | 0.435      | 9.28        | 2.09                   | 197.134                  |

Fig 8(a) shows the comparison of the delay and power dissipation of the two comparators, i.e. the conventional three-

stage comparator and the hysteresis-based comparator. The values are given in the table.

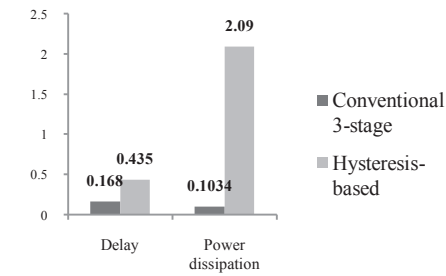


Fig.8 (a). Comparison of the delay and power dissipation of the two circuits

Fig 8(b) shows the comparison of the speed of the two comparators based on the values given in the table, found by simulation results.

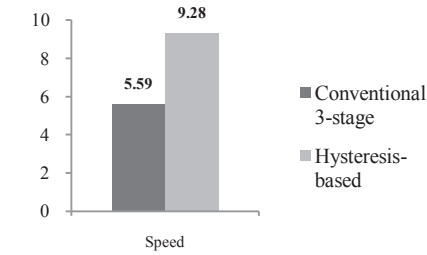


Fig. 8 (b). Comparison of the speed of the two circuits

## V. CONCLUSION

This paper presented the design of a hysteresis-based comparator, which is useful in cases of noisy inputs. The hysteresis introduced in the comparator, puts off noisy output to a great extent. As we know, the rms value of noise varies with many parameters (temperature, bandwidth and resistance). Hence, we need to be alert of how the noise may affect the comparator before designing it. The designed comparator showed higher speed of operation than the conventional 3-stage comparator. The area of the designed comparator is also considerable. Future work can be done to decrease the power dissipation to a further extent.

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